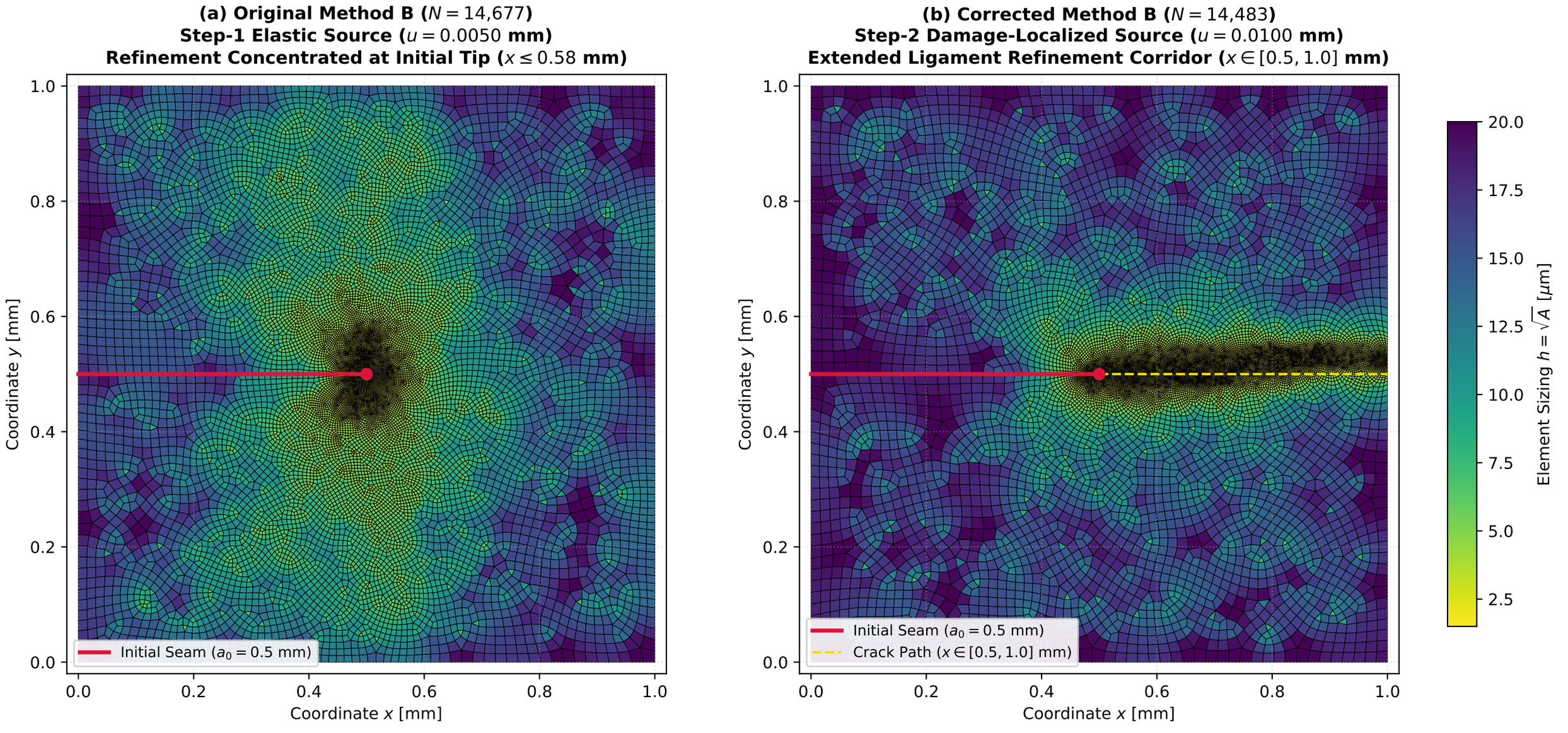

(a) Original Method B ($N = 14{,}677$)
Step-1 Elastic Source ($u = 0.0050$ mm)
Refinement Concentrated at Initial Tip ($x \leq 0.58$ mm)
(b) Corrected Method B ($N = 14{,}483$)
Step-2 Damage-Localized Source ($u = 0.0100$ mm)
Extended Ligament Refinement Corridor ($x \in [0.5, 1.0]$ mm)
Coordinate y [mm]
Coordinate x [mm]
0.0
0.2
0.4
0.6
0.8
1.0
Initial Seam ($a_0 = 0.5$ mm)
Crack Path ($x \in [0.5, 1.0]$ mm)
Element Sizing $h = \sqrt{A}$ [μm]
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0